

Machine learning for agricultural yield prediction: a case on sugar beet production in The Netherlands

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Introduction

Predicting crop yields is crucial in decision making for many stakeholders in the agri-food chain, like farmers, agronomists, commodity traders and policymakers (Basso and Liu 2019; Chipanshi et al. 2015; as cited in Paudel et al., 2020). However, predicting yield remains challenging because it depends on multiple crop-specific parameters, environmental conditions, and management factors. Traditional methods are increasingly being combined (field surveys, crop growth models, remote sensing, and statistical models) to improve yield forecasting and use machine learning as a promising tool to learn patterns from these multiple data sources. Machine learning offers key advantages, including the ability to model non-linear relationships between predictors, improve performance with more training data, and being robust to noisy data through regularization (Paudel et al., 2020). Predicting sugar beet yields in The Netherlands is crucial for market-decision-making in supply chains of many products available for consumers. As a result, this knowledge can aid supply chains in keeping high-sugar products; that increase the risk of obesity, diabetes type 2, cardiovascular disease, and cancer, on the supermarket shelves (Faruque et al., 2019; Wang et al., 2015; Yang et al., 2014; Epner et al., 2022). This paper implements a Random Forest model with an RMSE score of 7.99 using spatio-temporal open data from the European Commission's Joint Research Center (JRC) over the 11 provinces of The Netherlands from 1999 to 2018.

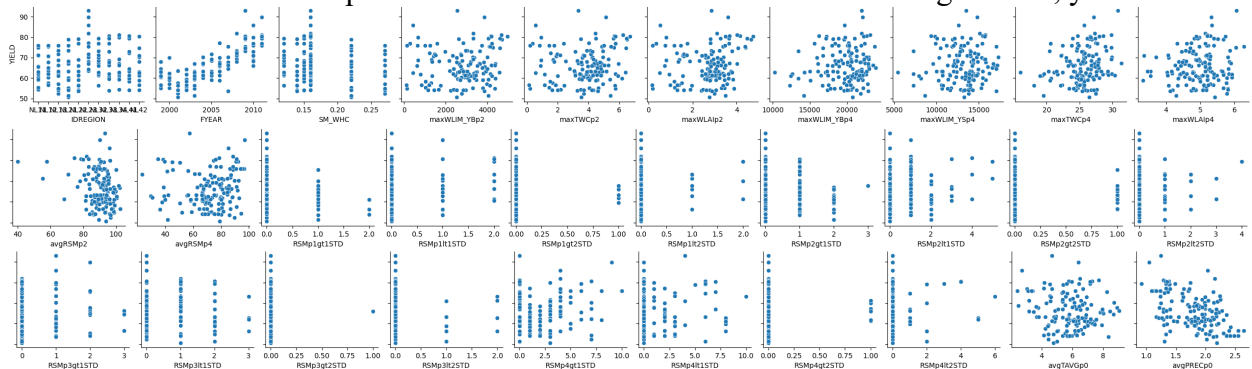
Methodology & Questions

2.1 Research question

To what extent do model types and dataset-reduction techniques affect the capabilities of sugar beet yield prediction in The Netherlands from 1999 to 2018?

2.2 Data exploration

Training data was used from 1999 to 2011 and testing data from 2012 to 2018. First, we wanted to visualize the relationships between the different estimators and the target value, yield.



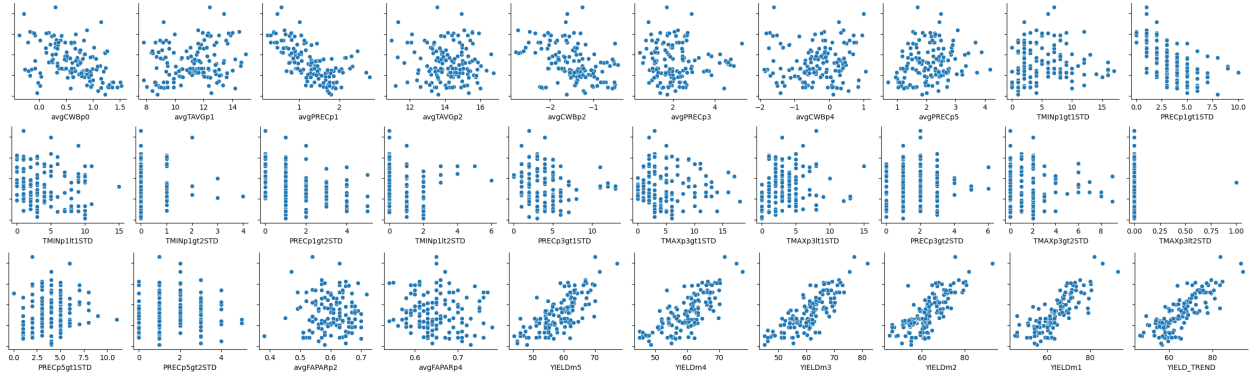


Figure 1: Target value (yield) and predictors plotted against each other. Yield is on the y-axis, the predictors on the x-axis.

The images above clearly indicate that there is no linear relationship between most of the predictors and our target value, yield. This indicated what models can be used to interpret the relationship between yield and the estimators.

However, we can observe some linear relationships between the estimators considering yield, as it is obvious that they are highly correlated with the value we are trying to predict.

As most relationships are non-linear, clearly linear regression models cannot capture this relationship. Therefore, non-linear regression models should be used. We decided to try PCA with Lasso and Ridge and Random Forest and a combination of those models.

Then, we plotted the distribution of the yield values that are used for the training of our models. The distribution of the target value is important as a distribution with extreme values, and outliers can impact model assumption and generalization performance.

As we can see, its distribution is normal, meaning that the distribution will not affect the models and the evaluations.

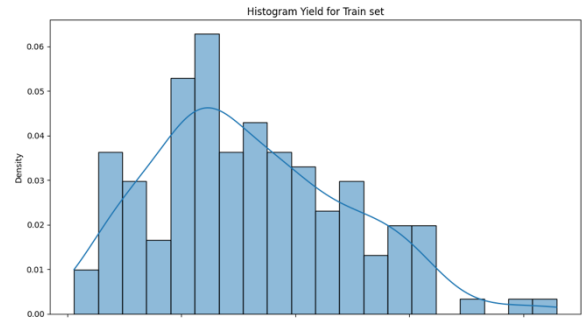


Figure 2: Distribution of the values of yield.

2.3 Pre-processing

Preprocessing is composed of sorting, converting categorical data to numerical data, and scaling the data. First, the training and test data were sorted by the year and id region columns because we perform cross-validation of our data using time series split, preserving chronological order and ensuring that validation data occurs after training data. We created dummy data for the column containing the ID of each region of the Netherlands because as a categorical column it could not be used otherwise. Finally, scaling was implemented using a Standard Scaler to scale the data properly.

2.4 Model choice and hyperparameter tuning

Several models and techniques were tested such as Lasso & Ridge Regression, ElasticNet, Support Vector Machines Regressors (SVR), Gradient Boosted Regressor, XGB Regression, Random Forest, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM) RNN, and

Convolution Neural Network (CNN). Additionally, 3 extra datasets were created using unsupervised and supervised dimensionality reduction techniques, and separation of data into static and temporal features: 1) Principal Component Analysis; 2) Features with non-zero coefficients in the lasso regression; and 3) Temporal data, defined in periods (p0-p5) for different weather and crop variables were converted into a 3D array where each sample has 6 rows defining the periods and 12 columns defining each temporal feature. As a result, a neural network can observe changes through time time.

Principal component dimensionality reduction (PCA) was implemented to lower the number of variables, decreasing the effect of dimensionality problems, since the training data has few samples (143 samples). Additionally, PCA reduces overfitting because of regularization, which means lower incidence of noise and multicollinearity (observed in features yieldm1, yieldm2 ... yieldm5).

The hyperparameters used for random forest used in our study are shown in the graph below.

	Parameter	Values
0	n_estimators	[10, 20, 50, 100]
1	max_depth	[2, 3, 4, 8]
2	min_samples_split	[0.5, 1, 2]
3	min_samples_leaf	[0.2, 0.5, 1]
4	max_features	[0.1, 0.3, 0.5]
5	max_samples	[0.5, 0.7, 0.9, None]
6	bootstrap	[True]

Figure 3: Hyperparameters used for random sampling.

The number of estimators decides the number of trees in the forest. Generally, larger values decrease the variance. Max depth determines the number of levels for each individual tree. Lower depth generally means less overfitting. The minimum number of splits specifies the minimum number of samples required to perform a split at an internal node. Lower values lead to more complex trees. The minimum number of samples decides the number of samples required to be in each node. Higher values lead to less overfitting as the model does not capture noise by creating small leaves. Max features specify the number of features considered when deciding for a split. Smaller values add more randomness, leading to better generalization. Max samples determine the fraction of training data used to train each tree. Small values lead to more randomness. Finally, bootstrap performs sampling with replacement of the data when building trees, increasing diversity.

Results

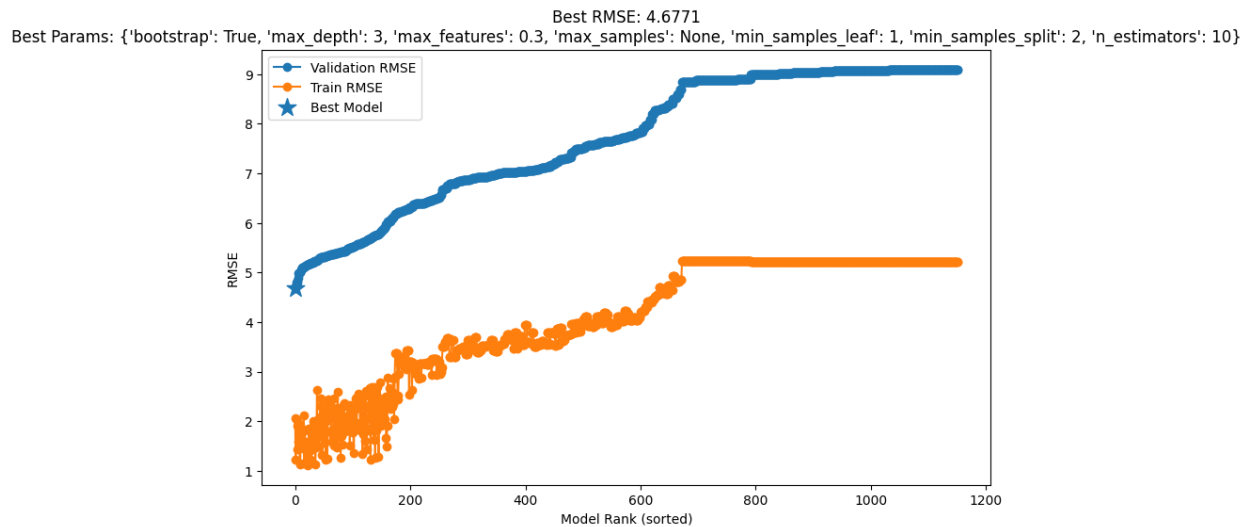


Figure 4: RMSE for models with all possible parameters for training data of the random forest model.

In the figure above, we can observe RMSE for the training data of the random forest model on the y-axis, and the index of each model sorted from high to low performance (low to high RMSE). The best model we ended up using was the one with a max depth of 3, max features of 0.3, no max samples , minimum samples per leaf of 1, minimum samples split of 2 and 10 estimators. We concluded that it was our best model because of its performance compared to the rest of the models trained.

Then, for the evaluation of all our models, root mean square error (RMSE) is used.

Model	Validation RMSE	Test RMSE
LSTM 3D Tensor	4.037	9.811
Random Forest	4.677	7.991
PCA + ElasticNet	4.946	9.031
PCA + Ridge	4.956	9.620
PCA + Lasso	4.959	9.599
XGBoost Regressor	4.983	8.306
Lasso	4.996	9.158
Ridge	5.044	8.970
SVR Linear	5.075	9.338
Random Forest w/ Lasso features	5.224	8.114
Gradient Boosting Regressor	5.244	8.443
SVR Sigmoid	5.297	8.825
SVR RBF	5.763	10.935
SVR Poly	6.223	24.042
PCA + RF	6.773	12.442
NN MLP	33.507	62.606
NN w/ Lasso features	39.575	73.031

CNN 3D Tensor	66.390	82.465
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Figure 5: Training RMSE for all the different models trained.

In the table above, the training error of each model can be observed. Random forest achieved the lowest test error (7.99), while its training error was also low (only LSTM using a 3D Tensor achieved lower training RMSE). As mentioned before, the non-linear models perform better due to the relationship between the predictors and the target value. Therefore, the difference between performance can be explained.

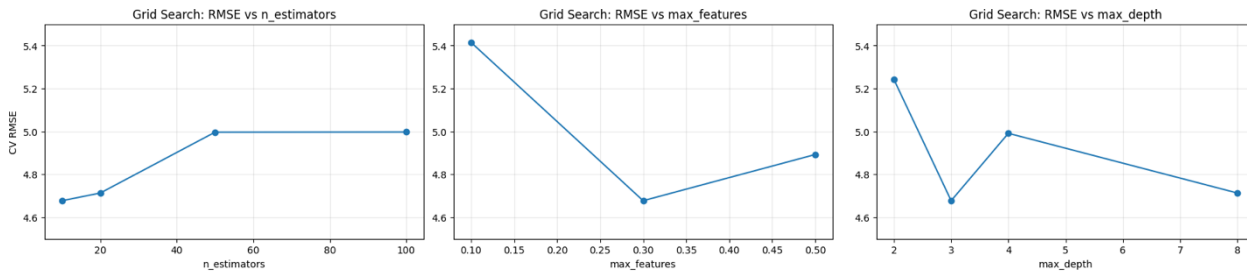


Figure 6: Random Forest best RMSE for hyperparameters

Diving into the effect of hyperparameters on the validation RMSE, the number of trees used in each forest shows a steady increase as the number increases. The max features, referring to the number of features to consider when looking for the best split, is lowest at 0.3. Finally, the maximum depth, referring to the maximum amount of decision levels in a tree, observed a minimum RMSE at 3 and an increase at 4.

Ethics

This research opens certain ethical, legal, and social dilemmas to point out and discuss. Firstly, ethical concerns arise when implementing these models expecting 100% accuracy and reliability. One might consider potential consequences of significant errors in the prediction of sugar beet yield for farmers' financial planning (eg. Buying in more fertilizers to potentially increase a yield whilst not necessary, or selling at lower prices when yield is expected to be high but turns out to be lower than expected). Therefore, clear communication of the models limitations must be enforced. Additionally, the extent of generalization of a model must be defined clearly since bias in the data collection process makes the model specific for specific contexts. For example, the current data is collected on largescale monocultures, meaning it may be biased towards this type of production, being difficult to predict in a mixed-cultivation or small scale production setting. In a broader context, ELSA advocates for research causing no harm. As mentioned in the introduction, one must question the societal benefit of the sugar agri-food industry as a whole. Even though our research doesn't directly cause harm to human health, the products that use this sugar beet yield do. This far fetched ethical dilemma falls onto the researcher and institution's moral compass and must be considered internally. In terms of legal concerns that must be highlighted, data acquisition and privacy are important, since competition between farms lead to insider information if not anonymized. Finally, social concerns as to who benefit from the results can lead to market imbalances, unfair competition and a widened gap between large agribusinesses and small scale farms.

Discussion

In general, the main advantage of random forest is its ability to capture non-linear relationships like the one between yield and the predictors in our data. Furthermore, it produces better results as it aggregates results by bootstrapping across many different decision trees and its random selection of features leads to reduced variance aiding the generalization of the model. Additionally, it is easy to implement as it does not require preprocessing of data, like scaling that other models require.

On the other hand, it can be computationally expensive especially when testing with several hyperparameters and a big number of folds. In our case, many parameters were considered and tested for tuning, and this led to long computation times (Figure 3). Also, random forest is susceptible to overfitting, especially when using a small dataset. In our case, this is something that should be considered as we observed that our training RMSE was low, almost around 5, while the test RMSE was 7.99. This difference highlights that the model overfitted the training data, leading to worse generalization due to higher variance.

Figure 6 highlights the importance of hyperparameter tuning as shallower trees, moderate number of trees and limited features can be seen performing better than models with hyperparameters that end up overfitting the data better.

Furthermore, linear models proved less useful for our dataset, as shown from the resulting training RMSE. This further confirms that models such as Lasso, Ridge, and ElasticNet fail to capture the nonlinearities present in the data, even though they can provide better interpretability and regularization.

Improvements to the research structure include implementing data augmentation techniques, reducing overfitting during training, and improving model generalization.

Conclusion and Outlook

Overall, Random Forest provided the most accurate predictions for the training dataset used for this study due to its predictive power of complex, nonlinear relationships between predictors and target values. However, the difference between the small training RMSE and the higher test RMSE indicates overfitting and weak generalization ability. This highlights that the model exhibits variance-related limitations.

To further improve performance, a further and more careful parameter tuning should be implemented. Hyperparameters such as maximum tree depth and minimum samples per leaf can improve generalization significantly, reducing variance in the model enhancing yield forecast. A solution can be using a range of values of hyperparameters instead of fixed. However, this is more computationally intense. Furthermore, dataset expansion should be considered, given that yields are affected by numerous underlying factors. Including additional relevant variables could strengthen the model predictability and improve its generalization ability. Implementing timeseries data correctly into Random Forest models could provide improved results since it cannot observe timeseries in the current input data structure.

Appendix

References

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Appendix

AI Usage Statement

AI tools were used to assist with code refinement, debugging, and clarification of methodological concepts. All analytical decisions, model implementation, and interpretation of results were conducted and verified independently by the authors. Code was generated to implement LSTM and CNN separating the temporal and static features.

Appendix A: Model reports

Version/Submission	Score	Notes
V1	8.28	Ridge with GridSearch with TimeSeries Split
V1.1	8.2	Lasso with GridSearch with TimeSeries Split
V1.2	8.3	PCA, then Lasso with GridSearch with TimeSeries Split
V1.3	8.1	PCA, then Ridge with GridSearch
V1.4	7.82	Random Forest with hyperparameter tuning and GridSearch with TimeSeries Split

V1.5	7.94	Best parameters of random forest, then stacking with lasso (alpha = 10, max_iterations = 10000)
V1.6	7.90	Best parameters of random forest, then stacking with lasso (alpha = 13, max_iterations = 10000)
V1.7	8.9	ElasticNet to combine Lasso and Ridge

Version/Submission	Score	Notes
lasso	9.14	Best alpha: 0.19306977288832497 Non-zero coefficients: 69
ridge	8.78	Best alpha: 0.0006866488450042998 Non-zero coefficients: 64
Svr linear	9.34	Best parameters: {'C': 10} Best score: - 5.03556034467562
Svr poly	10.80	Best parameters: {'C': 10, 'degree': 2} Best score: - 5.16551451639329
Svr rbf		Best parameters: {'C': 1000, 'gamma': 0.01} Best score: - 5.135679278211472
Svr sigmoid	9.46	Best parameters: {'C': 1000, 'gamma': 0.01} Best score: - 5.034903420584362
NN using lasso features	9.37	64 neuron and 1 neuron
Gradient boosting	8.44	Best parameters: {'alpha': 0.001, 'ccp_alpha': 0.0, 'max_depth': 2, 'n_estimators': 100, 'random_state': 42} Best score: - 5.242866998337822
Lstm using 3d time-aware data	14.07	
CNN using using 3d time-aware data	8.36	
Gradient Boosting using lasso features	8.55	
XGB	8.22	
	All above using Minmaxscaler	

	All bellow using Standard Scaler	
PCA + Random Forest	12.44	Best parameters: {'pca__n_components': 10, 'rf__bootstrap': True, 'rf__max_depth': 3, 'rf__max_features': 0.3, 'rf__max_samples': 0.7, 'rf__min_samples_leaf': 1, 'rf__min_samples_split': 2, 'rf__n_estimators': 10} Best score: - 6.772968900547636
Random forest with lasso features	7.66	
Random forest with whole dataset	7.99	Best params: {'bootstrap': True, 'max_depth': 3, 'max_features': 0.3, 'max_samples': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 10} Best RMSE: 4.677062118999499